FINAL PROJECT REPORT

Forecasting ESG Impact on Stock Prices

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ABSTRACT

This project investigates whether Environmental, Social, and Governance (ESG) factors improve the predictability of daily stock returns. Using a dataset that provides daily price movements and static ESG scores for publicly listed companies, we evaluate the effectiveness of ARIMA, VAR, and LSTM models on three representative firms with low, medium, and high ESG ratings. Our analysis shows that static ESG values do not contribute meaningful predictive information for daily forecasting. Classical ARIMA models perform comparably to LSTM, while VAR models reveal expected return–volatility dynamics. The findings align with prior literature suggesting that ESG influences long-term financial performance but provides limited short-term predictive power without time-varying ESG data.

1 INTRODUCTION

ESG investing has expanded rapidly as investors incorporate ethical and sustainability considerations into financial decision-making. This has raised important questions regarding whether ESG performance influences market behavior, and whether ESG-enhanced models can improve the forecasting of stock returns. In this project, we explore whether combining ESG scores with time-series methods yields better predictive performance compared to classical approaches in financial forecasting.

We benchmark several models—ARIMA, ARIMAX, Vector Autoregression (VAR), and LSTM—using daily stock data from three companies selected based on differing ESG scores. The core motivation is to determine whether ESG characteristics, even in static form, meaningfully enhance forecasting accuracy or structural insights.

2 PROBLEM STATEMENT

The central question is whether ESG characteristics contribute to forecasting daily stock returns. Specifically, we aim to determine:

(1) whether ESG data in this dataset varies in a way that supports time-series modelling,

(2) whether model performance differs across companies with different ESG scores, and

(3) whether deep learning models capitalize on any ESG-related information beyond traditional statistical models. Because daily returns are noisy and financial markets exhibit limited predictability, the study also evaluates how model behavior aligns with known characteristics of financial time-series data.

3 DATA DESCRIPTION

We use a single dataset that includes daily closing prices (2017–2025) and static ESG scores. For each company, ESG is provided as a *single constant value*. This key characteristic affects all modelling decisions. While the price series is rich and suitable for dynamic modeling, ESG cannot serve as a time-varying driver of returns. As a result, ESG can only be used in a cross-sectional manner rather than as a dynamic regressor. From the dataset, three companies are selected to represent low (Steel Partners Holdings L.P.), medium (C.H. Robinson Worldwide Inc.), and high ESG scores (Highway Holdings Ltd.), ensuring diversity in ESG characteristics while keeping the analysis tractable.

Preprocessing steps include computing log returns, creating lag features, generating 7-day and 30-day rolling volatilities, and cleaning the time index with business-day reindexing. For LSTM, sliding-window sequences and feature normalization are applied.

Fig 1: i) Price vs time plots showing long-term trends. ii) Log-return plots capturing day-to-day movement and volatility spikes

Fig 2: Rolling volatility and ESG over time (static for the dataset)

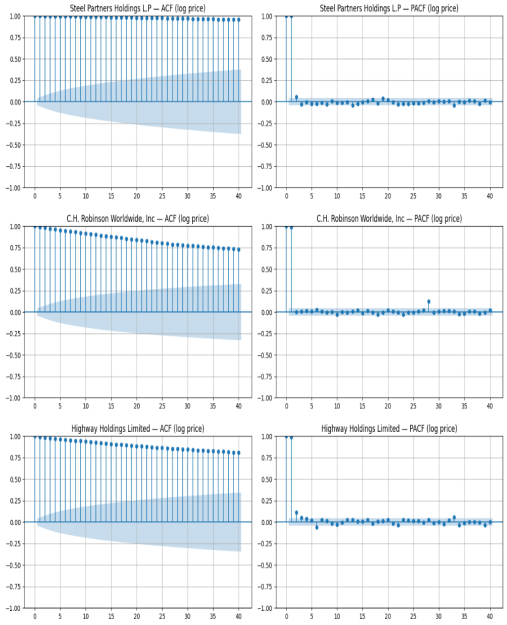


Fig 3: ACF and PACF plots

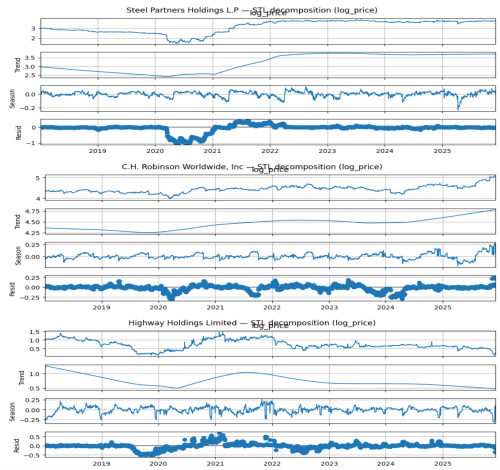
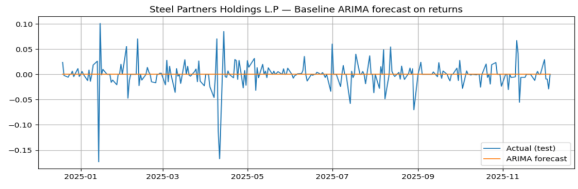


Fig 4: STL decomposition

4 METHODOLOGY

4.1 ARIMA and ARIMAX Models

ARIMA models serve as a baseline and are selected using AIC criteria. ARIMAX models incorporate ESG as an exogenous regressor; however, because ESG is constant over time, ARIMAX effectively collapses to ARIMA, offering no additional signal.



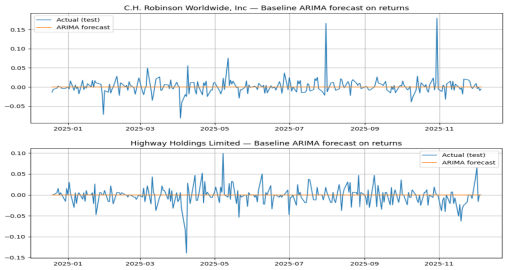


Fig 5: ARIMA models

4.2 VAR Models

VAR models allow joint modelling of returns and volatility. Impulse-response functions reveal how shocks propagate through the system and whether return–volatility interactions differ across ESG categories.

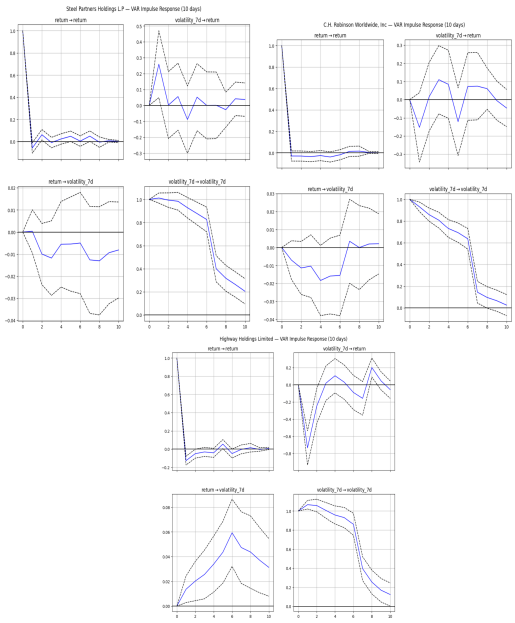
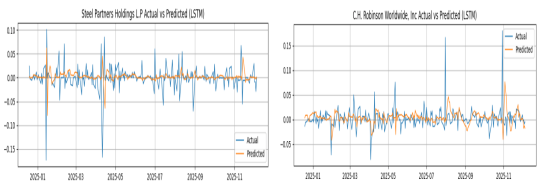


Fig 6: VAR Modelling

4.3 LSTM Models

A compact LSTM architecture is trained to predict next-day returns based on windows of past values and engineered features. While capable of capturing nonlinear patterns, its performance depends critically on the quality and richness of input signals.



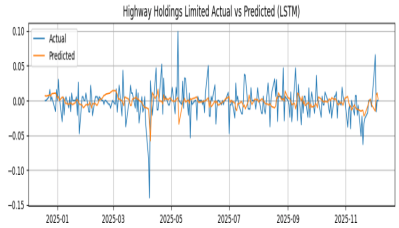


Fig 7: LSTM Models

4.4 Evaluation

Performance metrics include MAE, RMSE, MAPE, and directional accuracy. Diebold–Mariano tests provide statistical comparison between classical and deep learning forecasts.

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--- Company: Steel Partners Holdings L.P ---
  DM test (ARIMA vs LSTM): stat=-1.5368, p=0.1244, n=252

--- Company: C.H. Robinson Worldwide, Inc ---
  DM test (ARIMA vs LSTM): stat=-2.7657, p=0.0057, n=252

--- Company: Highway Holdings Limited ---
  DM test (ARIMA vs LSTM): stat=-0.6014, p=0.5476, n=252
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Fig 8: ARIMA vs LSTM : Diebold–Mariano test

5 RESULTS

Across all companies, ARIMA and LSTM models produce similar accuracy. For Steel Partners, ARIMA slightly outperforms LSTM; for CHRW, ARIMA is statistically superior based on the Diebold–Mariano test; for Highway Holdings, the models are statistically indistinguishable. VAR analysis shows that volatility shocks persist longer than return shocks, reflecting standard financial behavior.

Importantly, ESG does not influence forecasting performance. Since ESG is static, it cannot provide dynamic information for models such as ARIMAX or LSTM. Cross-sectional comparisons between the three companies also show no significant relationship between ESG levels and average returns or volatility.

Fig 9: Actual vs ARIMA vs LSTM

6 INTERPRETATION

The results demonstrate that static ESG scores do not enhance daily stock-return forecasting. This is consistent with financial theory: high-frequency market movements are driven by news, liquidity, and investor sentiment rather than slow-moving ESG indicators. The inability of LSTM to outperform ARIMA across all companies further underscores the low signal-to-noise ratio inherent in daily returns.

6.1 Literature Comparison

Earlier studies generally report that ESG influences *long-term* performance, risk, or valuation, especially when ESG data is updated periodically. Our results differ because the dataset provides only static ESG scores. Therefore, our findings do not contradict the literature; instead, they reinforce a key insight: ESG must be time-varying to influence short-term forecasts. Without temporal changes in ESG, models cannot capture any dynamic ESG-return relationship.

7 LIMITATIONS AND FUTURE WORK

This study is limited primarily by the static nature of the ESG data, which prevents meaningful time-series modeling of ESG effects. The analysis also focuses on only three companies, restricting cross-sectional insights. Future work should use time-varying ESG datasets, such as monthly ESG ratings, controversy scores, or news sentiment indices. Incorporating more firms and richer features would allow for more robust causal inference and forecasting.

8 CONCLUSION

This project develops a full forecasting pipeline, applying classical and deep learning techniques to assess whether ESG contributes predictive value for daily stock returns. The findings show that ESG, when static, does not improve model performance. ARIMA and LSTM models behave as expected for noisy financial data, and VAR models capture standard return–volatility interactions. The project highlights that meaningful ESG forecasting requires datasets where ESG varies over time, reaffirming lessons from existing financial literature.

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